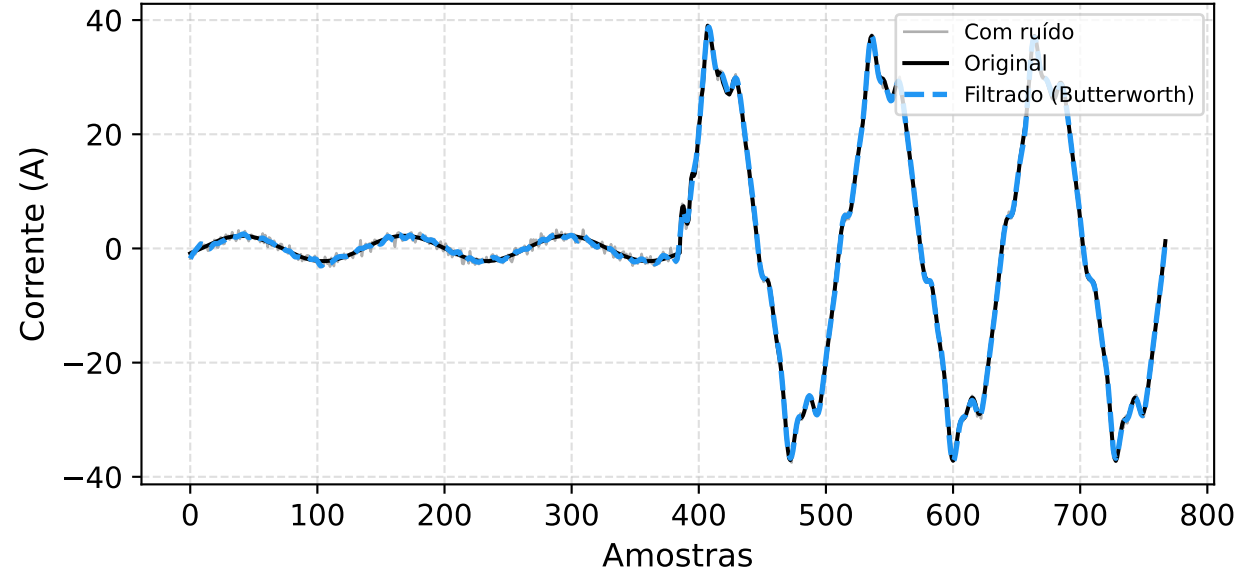
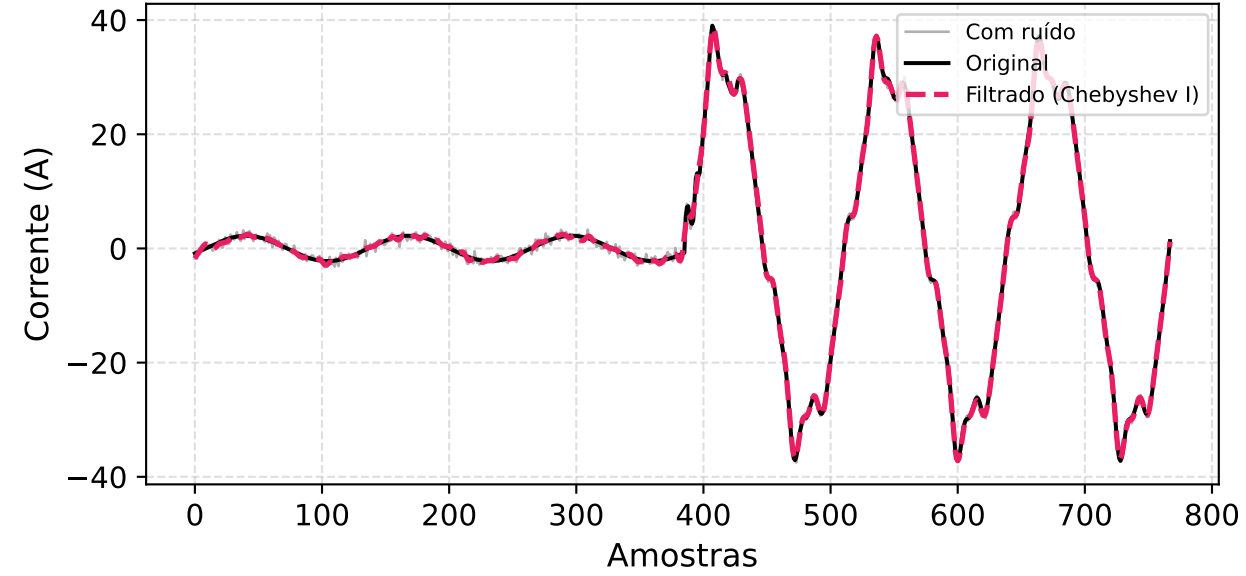


Filtragem IIR — Melhor Configuração por Aproximação (Fase Zero)

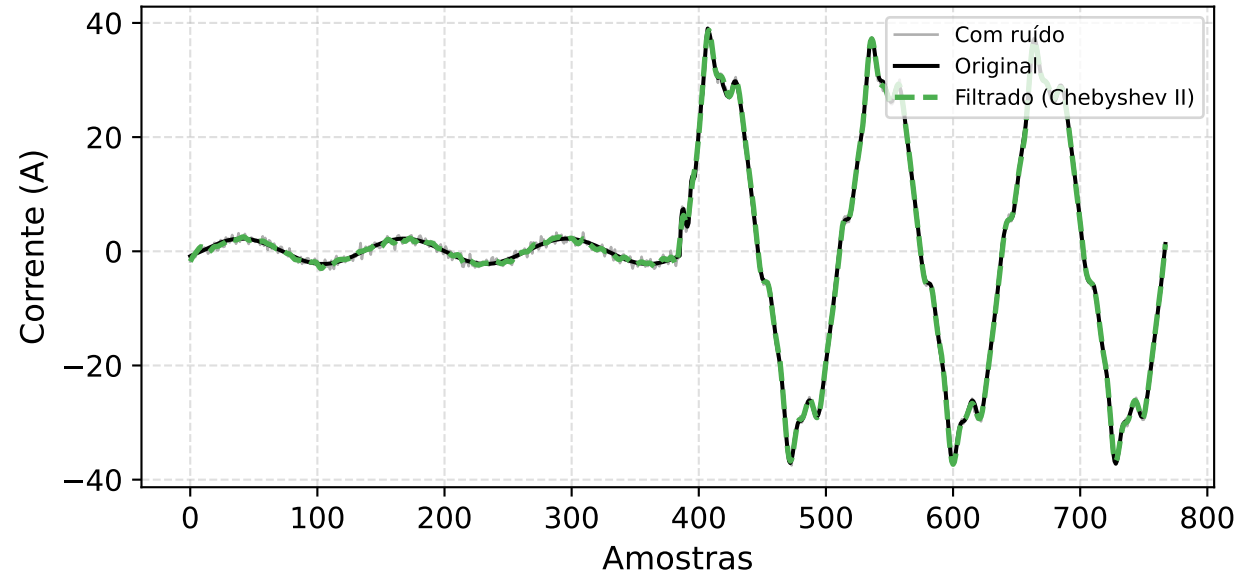
Butterworth — Ordem 3 | $f_c = 1100$ Hz
RMSE = 0.3210 | Δ SNR = 5.4 dB



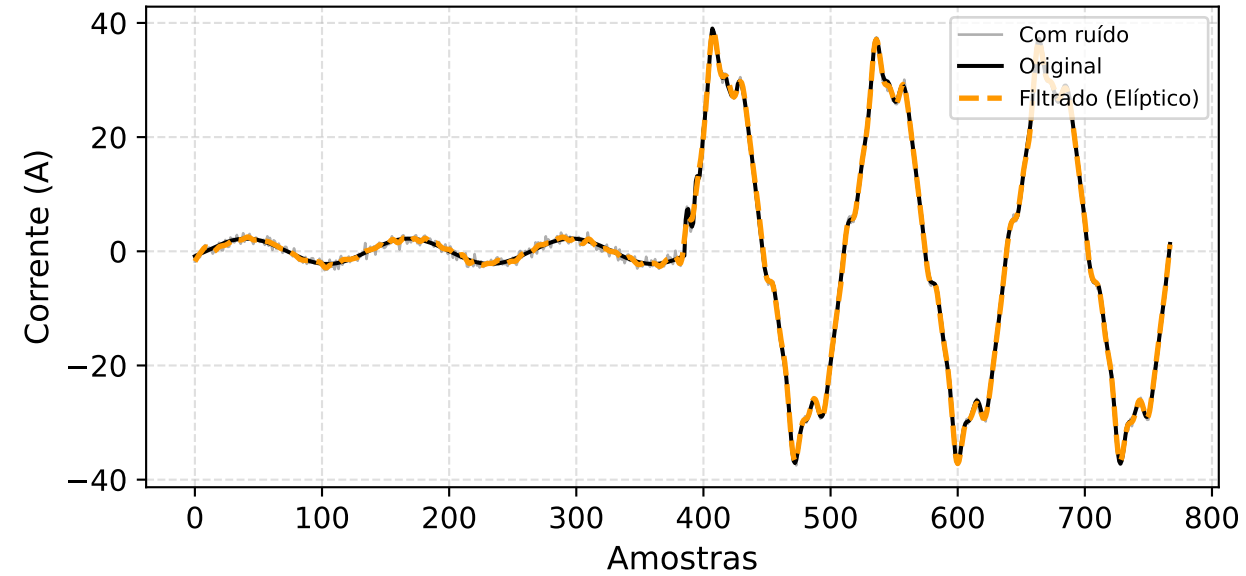
Chebyshev I — Ordem 3 | $f_c = 700$ Hz
RMSE = 0.3260 | Δ SNR = 5.2 dB



Chebyshev II ★ — Ordem 3 | $f_c = 2700$ Hz
RMSE = 0.3206 | Δ SNR = 5.4 dB



Elíptico — Ordem 3 | $f_c = 700$ Hz
RMSE = 0.3262 | Δ SNR = 5.2 dB



Erro residual (filtrado – referência) — zoom no transitório do chaveamento (amostras 300-460)

